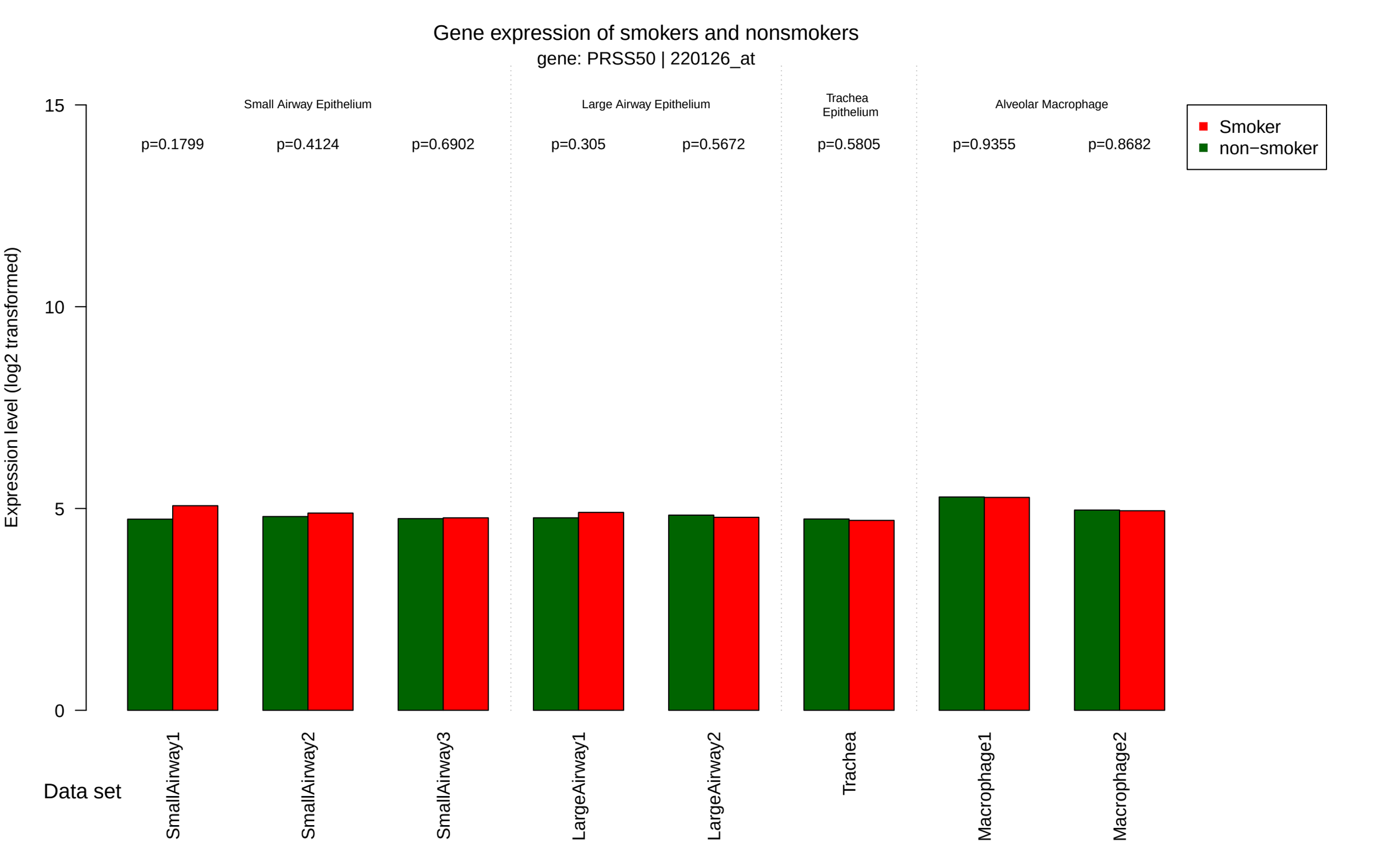
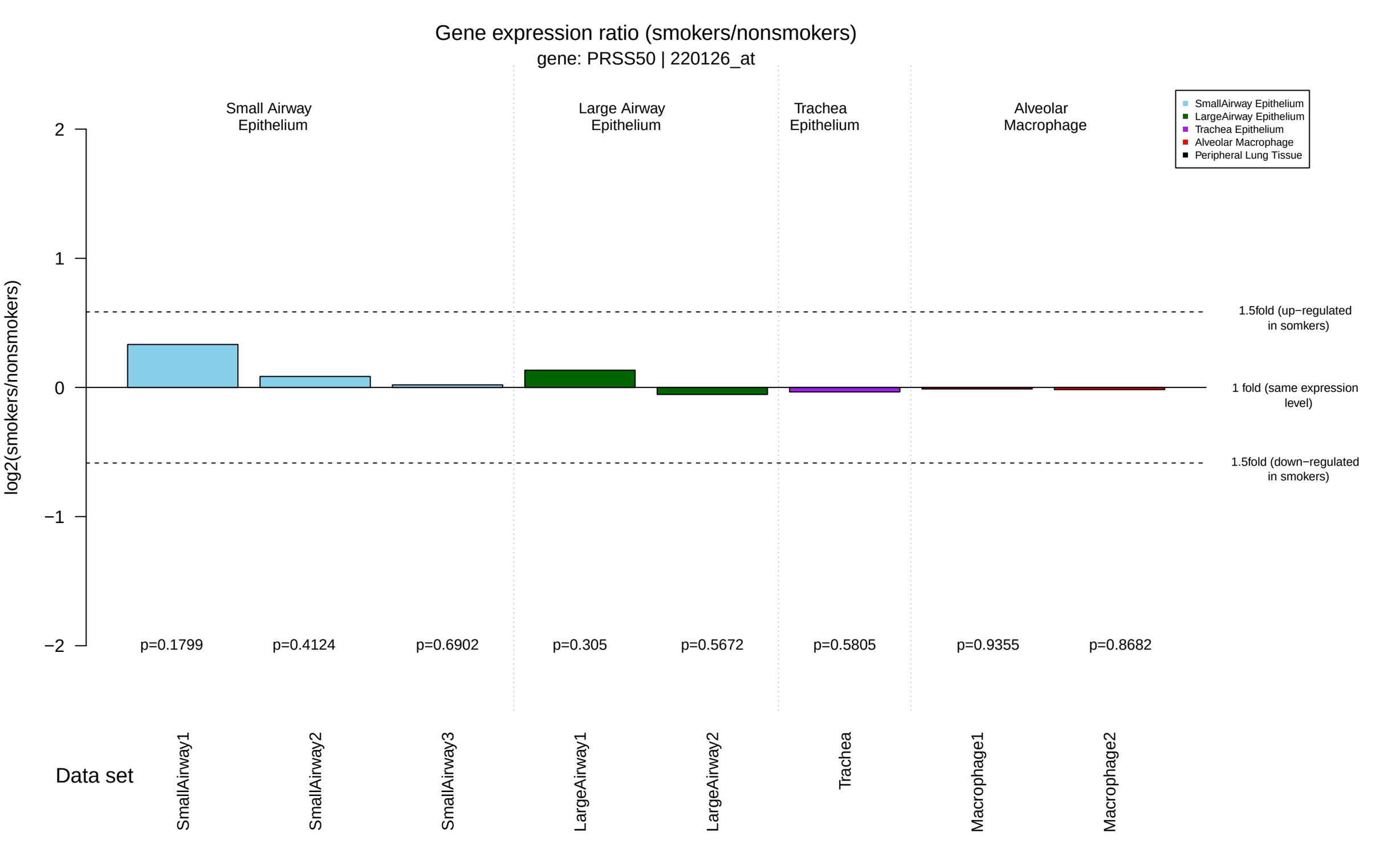
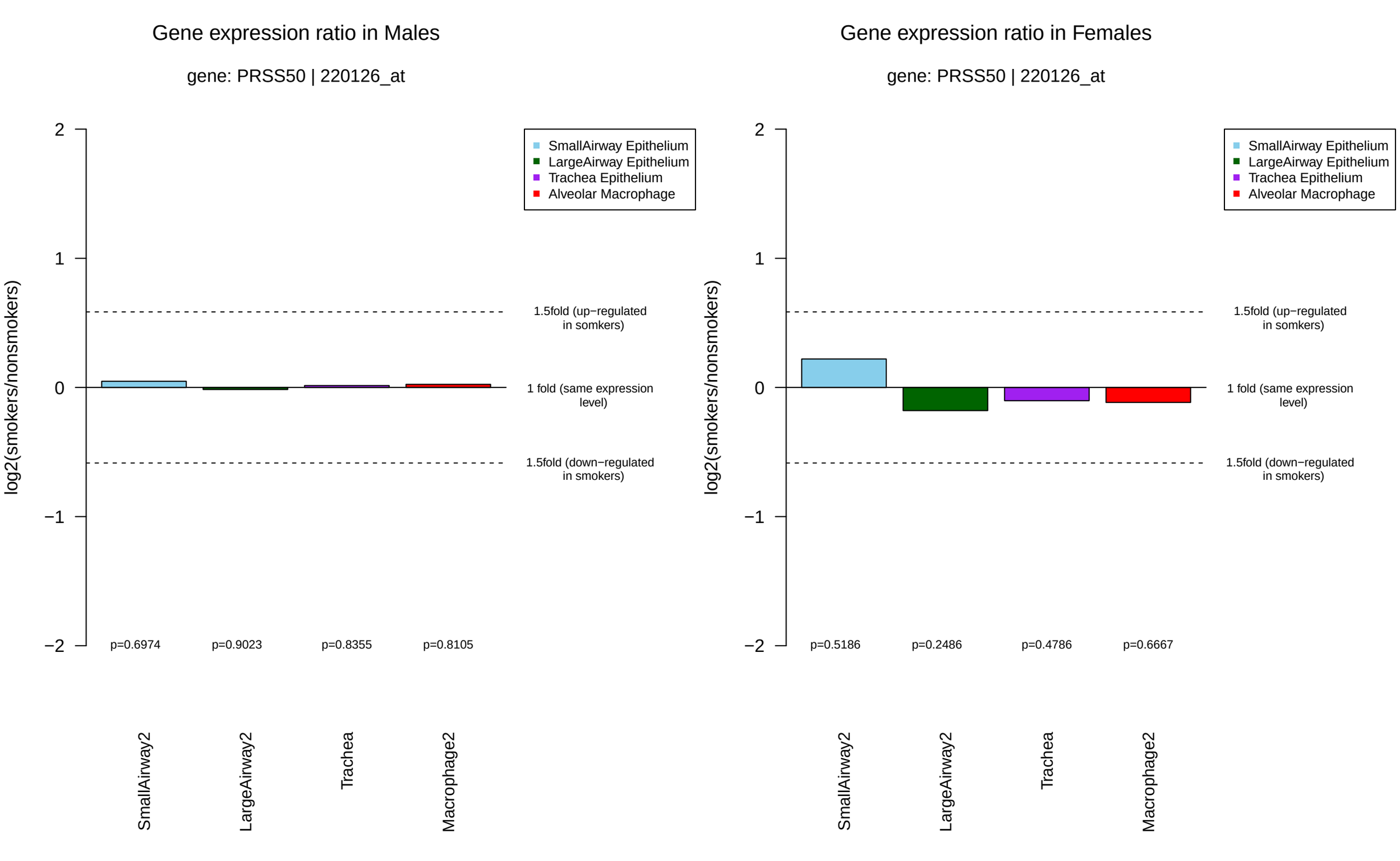
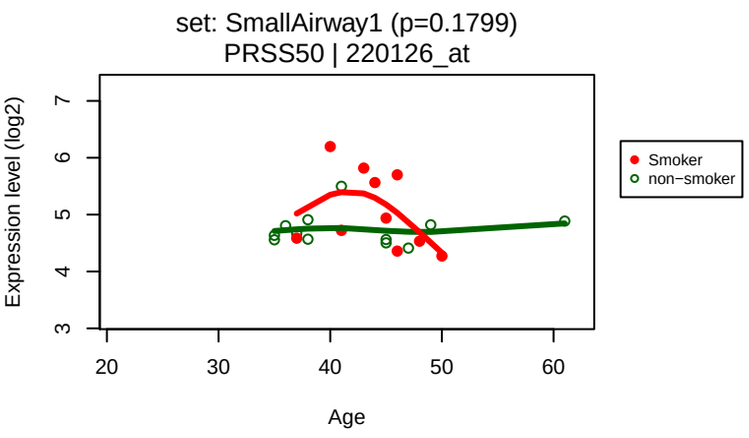
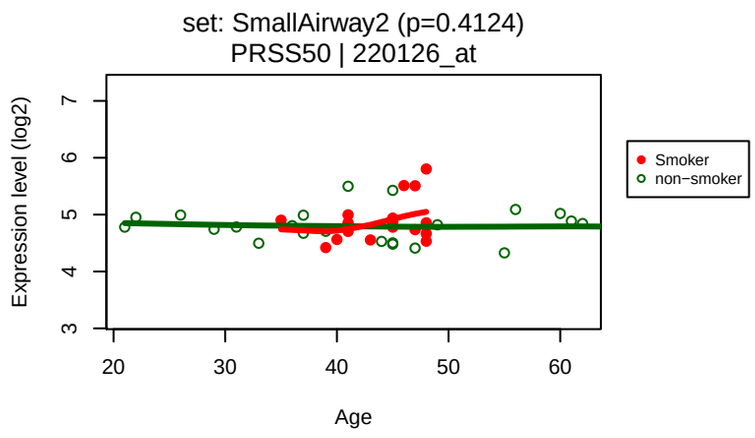
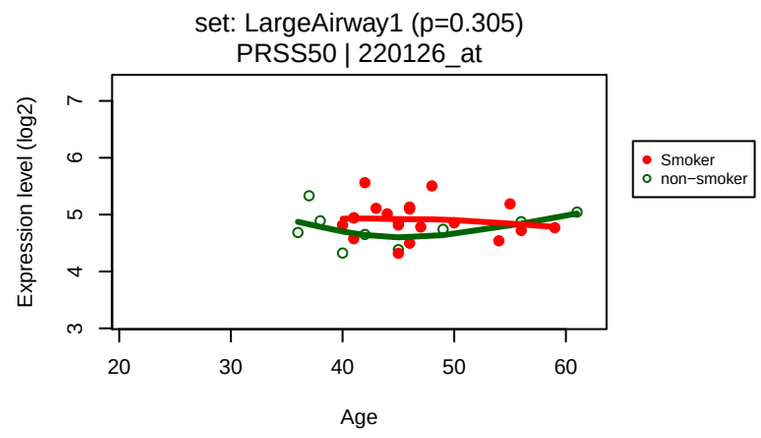

Gene expression of smokers and nonsmokers
gene: PRSS50 | 220126_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1799
p=0.4124
p=0.6902
p=0.305
p=0.5672
p=0.5805
p=0.9355
p=0.8682
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

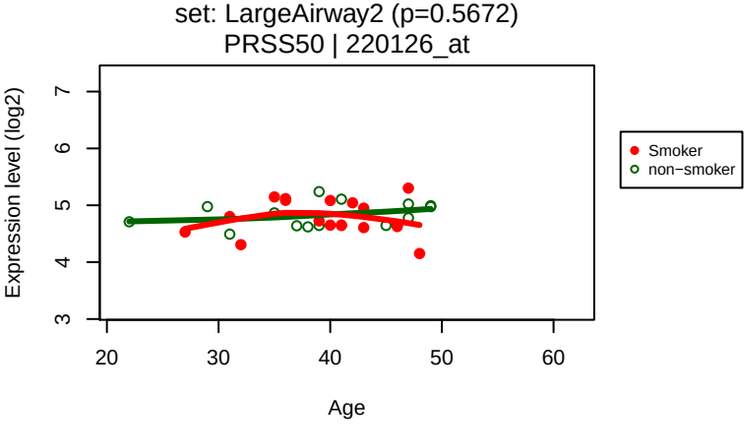

Gene expression ratio (smokers/nonsmokers)
gene: PRSS50 | 220126_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1799
p=0.4124
p=0.6902
p=0.305
p=0.5672
p=0.5805
p=0.9355
p=0.8682
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

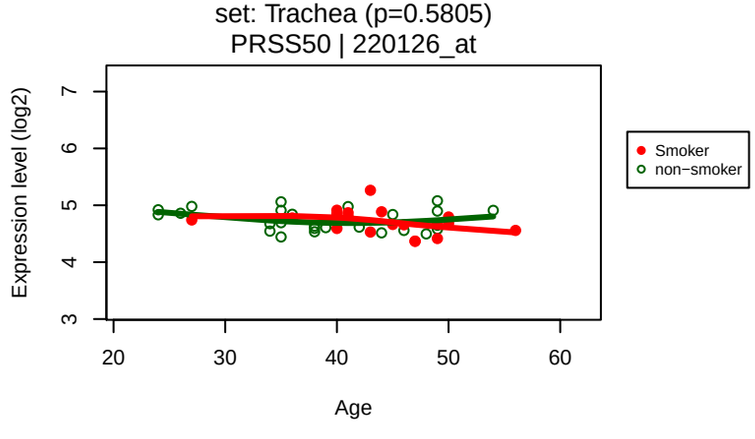

Gene expression ratio in Males
gene: PRSS50 | 220126_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6974
p=0.9023
p=0.8355
p=0.8105
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PRSS50 | 220126_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5186
p=0.2486
p=0.4786
p=0.6667
SmallAirway2
LargeAirway2
Trachea
Macrophage2

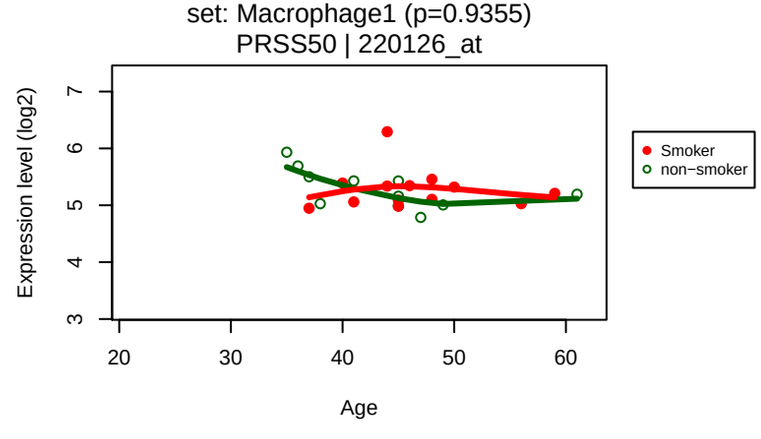

set: SmallAirway1 (p=0.1799)
PRSS50 | 220126_at
Expression level (log2)
Age
Smoker
non−smoker

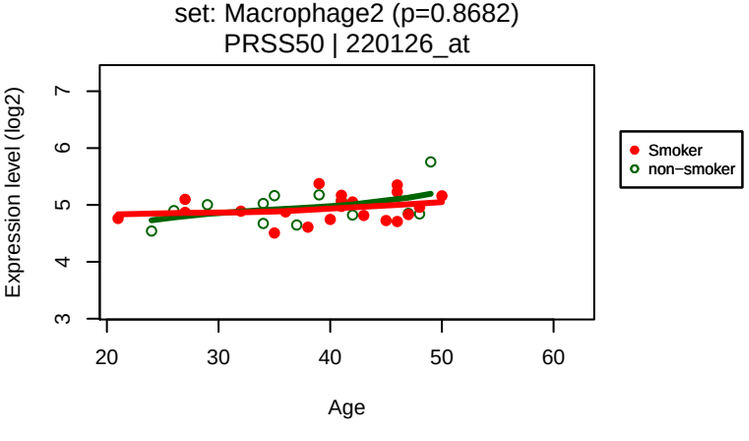

set: SmallAirway2 (p=0.4124)
PRSS50 | 220126_at
Expression level (log2)
Age
Smoker
non−smoker

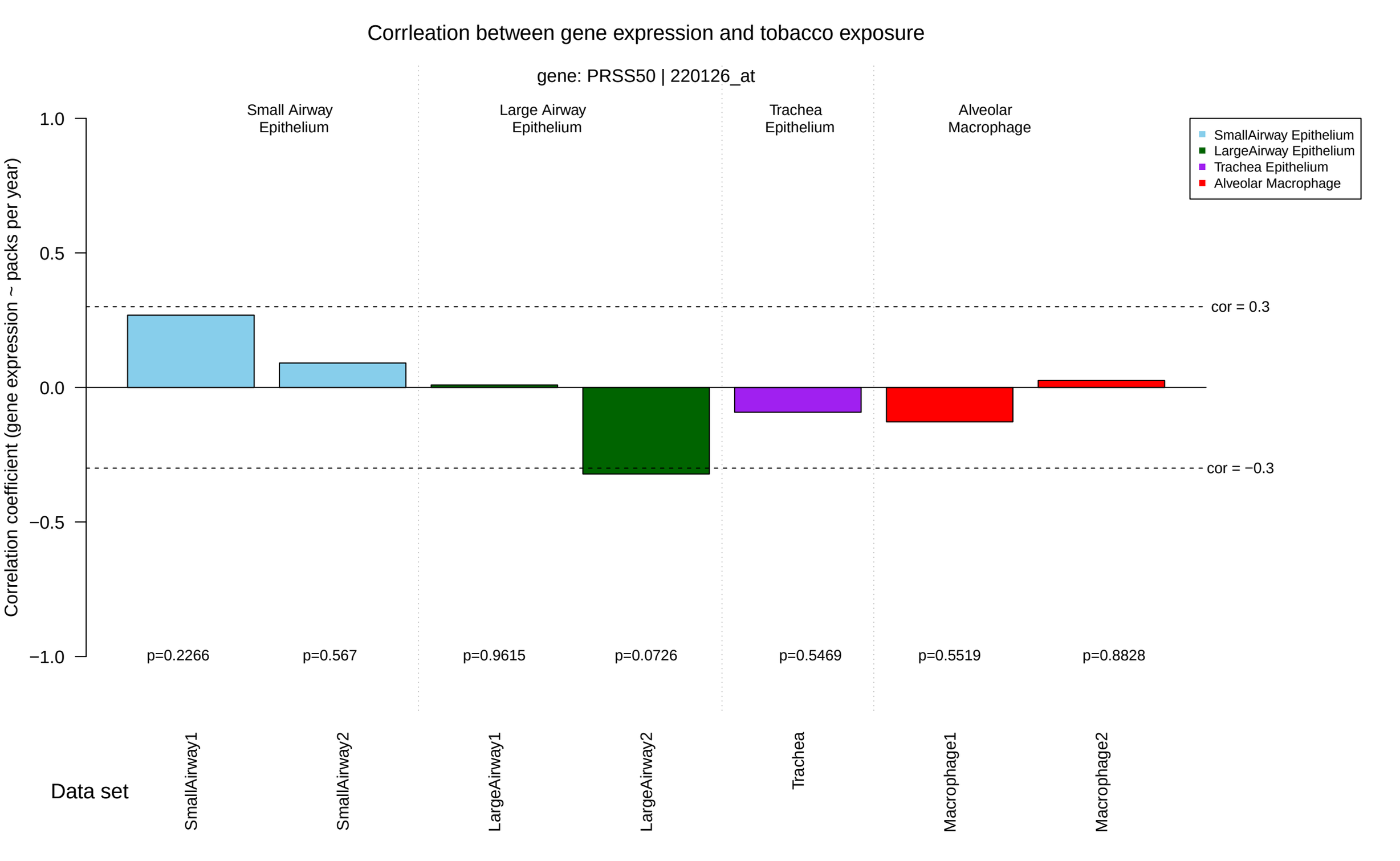

set: LargeAirway1 (p=0.305)
PRSS50 | 220126_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5672)
PRSS50 | 220126_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5805)
PRSS50 | 220126_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9355)
PRSS50 | 220126_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8682)
PRSS50 | 220126_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PRSS50 | 220126_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2266
p=0.567
p=0.9615
p=0.0726
p=0.5469
p=0.5519
p=0.8828
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2